

# IBKR Trading Bot

## Deployment & Operations Manual

Options Trading Bot — IB Gateway · Telegram · Three-Strategy System

Covers: VPS Setup · IB Gateway · Daily Operations  
Telegram Command Reference · Market-Hours Scheduling  
Troubleshooting · Emergency Procedures

*For paper trading use only until validated. Keep .env and credentials secure.*

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# Table of Contents

|                                            |    |
|--------------------------------------------|----|
| <b>1 VPS Requirements</b>                  | 3  |
| <b>2 One-Time Setup</b>                    | 3  |
| 2.1 System dependencies                    | 3  |
| 2.2 Clone and configure                    | 3  |
| 2.3 IB Gateway                             | 4  |
| 2.4 Bootstrap IV history                   | 4  |
| 2.5 systemd service                        | 4  |
| 2.6 First run                              | 5  |
| <b>3 Market-Hours Operation</b>            | 5  |
| <b>4 Shutdown &amp; Startup Procedures</b> | 6  |
| <b>5 Automating the Schedule</b>           | 6  |
| Option A — Cloud provider scheduler        | 6  |
| Option B — systemd timers (keep VM on)     | 7  |
| Option C — Always-on VPS                   | 7  |
| <b>6 Daily Trading Operations</b>          | 8  |
| <b>7 Scheduled Maintenance</b>             | 10 |
| <b>8 Troubleshooting</b>                   | 10 |
| <b>9 Emergency Procedures</b>              | 11 |
| <b>10 Quick Reference Card</b>             | 12 |

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## 1. VPS Requirements

### Minimum specification:

- 1 vCPU, 2 GB RAM, 20 GB disk
- Ubuntu 22.04 LTS (recommended)
- Static IP or fixed hostname

### Firewall — inbound ports to open:

| Port         | Protocol | Source       | Purpose                                   |
|--------------|----------|--------------|-------------------------------------------|
| 22           | TCP      | Your IP only | SSH access                                |
| (all others) | —        | CLOSED       | IB Gateway port 4002 must never be public |

■ ■ IB Gateway has no authentication. Port 4002 is bound to 127.0.0.1 only in docker-compose.yml — never change this or expose it publicly.

## 2. One-Time Setup

### 2.1 Install system dependencies

```
sudo apt update && sudo apt install -y \  
  python3.11 python3.11-venv python3-pip \  
  docker.io docker-compose-v2 git curl  
sudo systemctl enable --now docker  
sudo usermod -aG docker $USER  
# Log out and back in for the docker group to take effect
```

### 2.2 Clone, configure, and install

```
git clone https://github.com/rahuling/ibkr-trading-bot.git  
cd ibkr-trading-bot  
  
# Python virtual environment  
python3.11 -m venv venv  
source venv/bin/activate  
pip install -r requirements.txt  
  
# Directories  
mkdir -p data logs backups
```

### Configure environment variables:

```
cp .env.example .env  
nano .env
```

| Variable      | Example value | Notes                 |
|---------------|---------------|-----------------------|
| IBKR_USERNAME | myibkruser    | IBKR account username |
| IBKR_PASSWORD | *****         | IBKR account password |

| Variable                  | Example value | Notes                                        |
|---------------------------|---------------|----------------------------------------------|
| TRADING_MODE              | paper         | Keep 'paper' until validated on live account |
| TELEGRAM_BOT_TOKEN        | 123:ABC...    | From @BotFather                              |
| TELEGRAM_ALLOWED_USER_IDS | 123456789     | Your Telegram user ID (@userinfobot)         |

### Configure trading parameters:

```
cp config.yaml.example config.yaml
nano config.yaml

# Key settings to fill in:
# scanner.watchlist.core: [AAPL, MSFT, SPY, QQQ]
# scanner.watchlist.tactical: [AMD, META, TSLA]
# leap.momentum_watchlist: [NVDA, AMD, META]
# automation.level: 2          # human-approved (recommended)
```

## 2.3 Start IB Gateway

```
docker compose up -d

# Verify – wait 30-60 seconds for login
docker compose ps
docker compose logs -f    # look for 'Gateway Started'
```

**Note:** IB Gateway performs a mandatory restart at **11:59pm ET every night** (IBKR requirement). The bot detects the disconnect via heartbeat and auto-reconnects when the gateway comes back (~2 min). You will receive a Telegram alert on disconnect and a recovery notice on reconnect.

**Note:** IBKR permits only one active API session per account. Make sure TWS is not open on the same account simultaneously, or the gateway will fail to log in.

## 2.4 Bootstrap IV history (one-time, ~30 min)

The IVR scanner requires 252 trading days of historical IV per ticker before it can rank candidates. Run this once before your first scan:

```
source venv/bin/activate
python scripts/bootstrap_iv_history.py
```

Safe to re-run — uses INSERT OR IGNORE. After bootstrap, IV history is updated automatically at 4:15pm ET daily.

## 2.5 Install the systemd service

```
sudo cp systemd/trading-bot.service /etc/systemd/system/
sudo nano /etc/systemd/system/trading-bot.service

# Update these two lines:
User=ubuntu          # your VPS username
WorkingDirectory=/home/ubuntu/ibkr-trading-bot
```

The service file should contain:

```
[Unit]
```

```
Description=IBKR Trading Bot
After=network.target docker.service
Requires=docker.service

[Service]
Type=simple
User=ubuntu
WorkingDirectory=/home/ubuntu/ibkr-trading-bot
EnvironmentFile=/home/ubuntu/ibkr-trading-bot/.env
ExecStart=/home/ubuntu/ibkr-trading-bot/venv/bin/python bot/main.py
Restart=on-failure
RestartSec=10

[Install]
WantedBy=multi-user.target

sudo systemctl daemon-reload
sudo systemctl enable trading-bot    # auto-start on boot
```

## 2.6 First run — verify

```
sudo systemctl start trading-bot
sudo journalctl -u trading-bot -f    # watch startup
```

You should see in the logs:

```
Config loaded OK. Automation level: L2
IB Gateway connected. Net Liquidation: $...
Reconciliation: OK — DB and IBKR positions match
Scheduler started.
Bot started. Waiting for commands.
```

Open Telegram and send **/status** to confirm IB Gateway is connected.

## 3. Market-Hours Operation

### 3.1 What runs automatically

| ET Time          | Event                                                                              |
|------------------|------------------------------------------------------------------------------------|
| 9:30am           | Morning summary — portfolio value, open positions, expiry alerts, yesterday's P&L  |
| 9:45am           | Premium-selling scan (Core + Tactical) — proposals pushed to Telegram              |
| 3:00pm           | Afternoon premium-selling scan                                                     |
| 3:30pm           | EOD momentum scan — LEAP proposals                                                 |
| 4:15pm           | IV history update for all watchlist tickers                                        |
| Every 5 min      | Position monitor — live P&L/Greeks update, profit-target and strike-tested alerts  |
| Every 5 min      | Heartbeat — alerts on 2 consecutive IBKR disconnects; recovery notice on reconnect |
| 11:59pm          | IB Gateway mandatory restart (auto-reconnect, ~2 min downtime)                     |
| 1st of month 8am | Monthly P&L report pushed to Telegram                                              |

### 3.2 What requires your action

| Event                       | Your action                                                                  |
|-----------------------------|------------------------------------------------------------------------------|
| Trade proposal arrives      | <code>/approve &lt;id&gt;</code> or <code>/reject &lt;id&gt; [reason]</code> |
| Fill confirmation           | No action needed — informational                                             |
| Assignment alert            | CC proposal auto-generated — review + <code>/approve</code>                  |
| Strike tested alert         | Your CSP is near its strike — decide: hold, roll, or close                   |
| Profit target alert         | Bot suggests closing — use <code>/close &lt;id&gt;</code> to act             |
| Loss limit hit / Bot paused | Review positions, then <code>/resume</code> when satisfied                   |
| Heartbeat disconnect alert  | SSH to VPS, check docker compose ps, restart if needed                       |

## 4. Shutdown & Startup Procedures

### 4.1 Manual daily shutdown (after 4:30pm ET)

```
# Stop the bot
sudo systemctl stop trading-bot

# Stop IB Gateway
cd /home/ubuntu/ibkr-trading-bot
docker compose down

# Power off the VM
sudo shutdown -h now
```

Or as a one-liner:

```
sudo systemctl stop trading-bot && \  
docker compose -f /home/ubuntu/ibkr-trading-bot/docker-compose.yml down && \  
sudo shutdown -h now
```

## 4.2 Manual startup (before 9:20am ET)

Start the VM from your cloud provider's console, then SSH in:

```
cd /home/ubuntu/ibkr-trading-bot  
docker compose up -d  
sleep 60 # wait for IB Gateway login  
sudo systemctl start trading-bot  
# Confirm in Telegram  
# Send /status — should show IB Gateway: ■ Connected
```

## 5. Automating the Schedule

Since the VM is powered off, it cannot schedule its own startup. Choose one of the three options below.

### Option A — Cloud provider scheduler (recommended if using AWS/GCP)

Target window: **start at 9:15am ET, stop at 4:45pm ET, Mon–Fri.**

**AWS EC2** — EventBridge + Systems Manager (or Instance Scheduler):

```
# Start rule: 9:15am ET = 14:15 UTC (EDT) / 13:15 UTC (EST)
aws events put-rule \
  --schedule-expression 'cron(15 14 ? * MON-FRI *)' \
  --name ibkr-bot-start --state ENABLED

# Stop rule: 4:45pm ET = 21:45 UTC (EDT) / 20:45 UTC (EST)
aws events put-rule \
  --schedule-expression 'cron(45 21 ? * MON-FRI *)' \
  --name ibkr-bot-stop --state ENABLED
```

■ ■ US daylight saving time shifts the UTC offset between -4 (EDT, Mar–Nov) and -5 (EST, Nov–Mar). Adjust your UTC cron times at each DST change, or use AWS Instance Scheduler which handles DST automatically via the America/New\_York timezone.

**GCP Compute Engine:** VM Manager → Instance Schedules → create schedule with America/New\_York timezone.

**DigitalOcean / Hetzner / Linode:** No native scheduler — use Option B or Option C.

### Option B — systemd timers (keep VM always on, schedule bot only)

The VM runs 24/7 but the bot and gateway only run during market hours. Simpler than cloud scheduling and handles DST automatically.

```
# /etc/systemd/system/ibkr-start.service
[Unit]
Description=Start IBKR bot for market open
[Service]
Type=oneshot
User=root
WorkingDirectory=/home/ubuntu/ibkr-trading-bot
ExecStart=/usr/bin/docker compose up -d
ExecStartPost=/bin/sleep 60
ExecStartPost=/bin/systemctl start trading-bot
```

```
# /etc/systemd/system/ibkr-start.timer
[Unit]
Description=Start IBKR bot at market open
[Timer]
OnCalendar=Mon-Fri 09:15:00 America/New_York
Persistent=false
```



```
[Install]
WantedBy=timers.target

# /etc/systemd/system/ibkr-stop.service
[Unit]
Description=Stop IBKR bot after market close
[Service]
Type=oneshot
ExecStart=/bin/systemctl stop trading-bot
ExecStartPost=/usr/bin/docker compose \
    -f /home/ubuntu/ibkr-trading-bot/docker-compose.yml down

# /etc/systemd/system/ibkr-stop.timer
[Unit]
Description=Stop IBKR bot after market close
[Timer]
OnCalendar=Mon-Fri 16:30:00 America/New_York
Persistent=false
[Install]
WantedBy=timers.target

sudo systemctl daemon-reload
sudo systemctl enable --now ibkr-start.timer ibkr-stop.timer

# Verify
systemctl list-timers ibkr-*
```

**Note:** Requires systemd 242+. Ubuntu 22.04 ships systemd 249 — compatible.

### Option C — Always-on VPS (simplest)

For a \$5–6/month nano VPS, always-on costs less than the operational overhead of start/stop scheduling. With **systemctl enable trading-bot** already set, the bot starts automatically on every boot. The IB Gateway nightly restart is handled automatically.

## 6. Daily Trading Operations

### 6.1 Typical morning flow

- **9:30am** — Morning summary arrives. Check portfolio value, open positions, anything expiring this week.
- **9:45am** — Scan results arrive. 0–5 trade cards depending on IV conditions.
- **Review each card** — check underlying, strike, DTE, delta, IVR. Approve if it meets your criteria.
- **Fill confirmation** — arrives within seconds to a few minutes once submitted.

### 6.2 Approving and rejecting proposals

```
/approve ABC123          # approve – bot fetches fresh market data first
/reject ABC123 IV drop   # reject with optional reason
```

The bot fetches a **fresh** bid-ask mid at approval time — not the stale price from scan time. You'll see the actual submission price in the confirmation.

***Note:** Proposals expire at 3:55pm ET even if the TTL hasn't elapsed, so you can't accidentally submit an order after the close blackout window.*

### 6.3 Checking open positions

```
/positions
```

Shows all open positions with live P&L, delta ( $\Delta$ ), theta ( $\Theta$ ), and trade ID.

### 6.4 Rolling a CSP

When a CSP gets close to its strike you'll receive a **Strike Tested** alert. Check roll economics:

```
/roll <trade_id>

# Output shows: current strike, roll strike, close debit, new credit, net credit
# To execute the roll:
/close <trade_id>          # 1. close current position
/scan                      # 2. generate fresh proposals
/approve <new_proposal_id> # 3. enter the new position
```

### 6.5 The Wheel — assignment and covered call

When a CSP is assigned:

- You receive an **Assignment** alert with net cost basis (strike – premium collected)
- A **Covered Call proposal** is automatically generated and sent to Telegram
- Review it — check that the strike is above cost basis ('above cost basis' tag confirms profit if called away)
- **/approve <id>** to sell the covered call

If the CC proposal doesn't arrive (market was closed at assignment time):

```
/scan    # detects assigned position and generates CC proposal
```

### 6.6 Closing a position manually

```
/close <trade_id>          # limit order at mid (default)
/close <trade_id> bid      # 1 tick above mid – more urgent
/close <trade_id> market   # emergency only – warning alert sent first
```

If the close order doesn't fill, the bot reprices once (1 tick) before giving up. You'll be notified either way.

## 6.7 Wheel cycle P&L;

```
/wheel          # list recent cycles
/wheel <cycle_id> # full P&L for one cycle (CSP + CC combined)
```

## 6.8 Journal and analytics

```
/journal 30      # last 30 days of closed trades
/journal 7       # just this week
/analyze ivr     # win rate by IVR bucket
/analyze dte     # win rate by DTE bucket
/analyze strategy # by strategy
```

## 6.9 Risk dashboard

```
/risk
```

Shows bucket usage (Core/Tactical/Momentum), daily/weekly/monthly P&L; vs limits, PDT status.

## 6.10 Adjusting settings at runtime

```
# View all current settings
/config

# Adjust a parameter (changes take effect immediately, are audited in DB)
/setconfig risk.min_ivr_core 35
/setconfig risk.daily_loss_limit_pct 0.02
/setconfig automation.level 1      # alerts-only mode

# Manage watchlist at runtime
/addticker GOOGL core
/addticker MARA momentum
/removeticker COIN tactical
/watchlist
```

**Note:** Watchlist changes made via `/addticker` and `/removeticker` are in-memory only — they reset on bot restart. To persist them, update `config.yaml`.

## 7. Scheduled Maintenance

### 7.1 Log monitoring

```
# Live stream
sudo journalctl -u trading-bot -f

# Last 100 lines
sudo journalctl -u trading-bot -n 100

# Today only
sudo journalctl -u trading-bot --since today

# Rotating file log (also written here)
tail -f /home/ubuntu/ibkr-trading-bot/logs/bot.log
```

### 7.2 Database backup

```
cd /home/ubuntu/ibkr-trading-bot
bash scripts/backup.sh
ls backups/

# Schedule weekly backup (add to crontab)
crontab -e

# Add this line:
0 20 * * 5 /home/ubuntu/ibkr-trading-bot/scripts/backup.sh
```

### 7.3 Updating the bot

```
sudo systemctl stop trading-bot
cd /home/ubuntu/ibkr-trading-bot
git pull
source venv/bin/activate
pip install -r requirements.txt # only if requirements changed
sudo systemctl start trading-bot
```

## 8. Troubleshooting

### 8.1 IB Gateway disconnected / heartbeat alert

```
docker compose ps          # check container status
docker compose logs --tail 50 # look for login errors
docker compose restart      # restart gateway
```

Common causes:

- Wrong credentials in `.env`
- TWS or another API session open on the same IBKR account simultaneously
- IB Gateway nightly restart at 11:59pm ET got stuck — docker compose restart fixes this

The bot reconnects automatically once the gateway is available. Confirm with `/status`.

### 8.2 Reconciliation mismatch on startup

```
/reconcile
```

| Mismatch type      | Cause                            | Fix                                                            |
|--------------------|----------------------------------|----------------------------------------------------------------|
| In DB but not IBKR | Position closed in TWS / expired | Use <code>/close &lt;id&gt;</code> to mark it closed in the DB |
| In IBKR but not DB | Position opened manually in TWS  | Close it in TWS, or add it to DB manually                      |

### 8.3 Proposal expired before approval

```
/scan    # generates fresh proposals
```

### 8.4 Order submitted but no fill notification

```
/positions  # if it appears here, fill was recorded
# If not, check logs:
sudo journalctl -u trading-bot -n 200 | grep -E 'fill|order|reprice'
```

The bot reprices once after 5 minutes and cancels if still unfilled after 3 more minutes. You'll receive a notification either way.

### 8.5 Bot won't start

```
sudo journalctl -u trading-bot -n 50  # read the error
sudo systemctl status trading-bot
```

| Error                   | Fix                                                |
|-------------------------|----------------------------------------------------|
| Config validation error | Fix the offending value in config.yaml and restart |
| Database locked         | fuser data/trading.db — kill the locking process   |
| IB Gateway unreachable  | docker compose up -d first, then start the bot     |

### 8.6 Was my order filled during a crash?

On restart, the bot automatically runs **recover\_orphaned\_orders**: it checks pending\_submit orders against live IBKR open orders, and submitted orders against IBKR execution reports. If a fill happened during the crash it is recovered automatically and you'll receive a fill notification. Then run **/reconcile** to confirm.

## 9. Emergency Procedures

### 9.1 Immediately stop all trading

```
/pause    # blocks all new proposals and order submission
           # existing open positions are unaffected
/resume    # when ready to trade again
```

### 9.2 Emergency close a position

```
/close <trade_id> market  # market order – warning alert sent first
```

■ Only use market orders if you need out immediately regardless of price. Always try limit first.

### 9.3 Close position in TWS (bypass the bot entirely)

Close the position directly in TWS, then:

```
/reconcile # bot detects the mismatch and alerts you
```

The bot won't silently update the DB from TWS closes. To mark the trade closed manually:

```
sqlite3 /home/ubuntu/ibkr-trading-bot/data/trading.db \  
"UPDATE trades SET status='closed', exit_reason='manual_tws',  
exit_date=datetime('now') WHERE trade_id='<id>';"
```

## 9.4 Kill the bot without systemd

```
pkill -f 'python bot/main.py'
```

## 10. Quick Reference Card

### Telegram commands

| Command                    | Description                                               |
|----------------------------|-----------------------------------------------------------|
| /status                    | IB Gateway connection, account, balance, automation level |
| /positions                 | Open positions with live P&L, delta, theta                |
| /risk                      | Capital allocation, bucket usage, loss limits, PDT status |
| /scan                      | Trigger manual scan (CSP, Spread, + pending CC proposals) |
| /approve <id>              | Approve proposal — fetches fresh price, submits order     |
| /reject <id> [reason]      | Reject proposal                                           |
| /close <id>                | Limit close at mid price                                  |
| /close <id> bid            | Urgent close — 1 tick above mid                           |
| /close <id> market         | Emergency market close                                    |
| /roll <id>                 | Show roll economics for an open CSP                       |
| /wheel [id]                | Wheel cycle P&L — omit id for recent cycle list           |
| /journal [days]            | Closed trade log (default 30 days)                        |
| /analyze [dim]             | Win rate by: ivr / dte / strategy / rule tag              |
| /config                    | Show current configuration                                |
| /setconfig <param> <value> | Adjust a parameter at runtime                             |
| /watchlist                 | Show all watchlists                                       |
| /addticker <T> <bucket>    | Add ticker to core / tactical / momentum                  |
| /removeticker <T> <bucket> | Remove ticker                                             |
| /pause                     | Halt scanning and order submission                        |
| /resume                    | Resume after pause                                        |
| /reconcile                 | Force DB ↔ IBKR position comparison                       |

### VPS / shell commands

| Command                               | Purpose                        |
|---------------------------------------|--------------------------------|
| sudo journalctl -u trading-bot -f     | Live log stream                |
| sudo systemctl restart trading-bot    | Restart bot                    |
| sudo systemctl stop/start trading-bot | Stop or start bot              |
| docker compose ps                     | Check gateway container status |
| docker compose restart                | Restart IB Gateway             |

| Command                             | Purpose         |
|-------------------------------------|-----------------|
| <code>docker compose logs -f</code> | IB Gateway logs |
| <code>bash scripts/backup.sh</code> | Backup database |

### Automation level reference

| Level                   | Behaviour                                                           |
|-------------------------|---------------------------------------------------------------------|
| L1 — Alerts only        | Bot scans and notifies; no order submission                         |
| L2 — Assisted (default) | Bot proposes, human approves via <code>/approve</code>              |
| L3 — Autonomous         | Bot executes without approval (except positions > 20% of portfolio) |